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'FOR INTERRUPT
Dim int_count As Byte
Dim en_count As Word
'MAIN LOGO AND DISPLAY INITIALISE
LcdInit 0
WaitMs 10
LcdCmdout LcdClear
LcdCmdout LcdLine1Home
Lcdout "Solar Charge 1.0"
LcdCmdout LcdLine2Home
Lcdout "SOLAR CHARGE.hex"
WaitMs 2000
' FOR ENERGY CALCULATIONS
Dim en_unit As Byte
Dim en_ten As Byte
Dim en_hun As Byte
Dim en_thou As Byte
Dim en_total As Word
Dim en_unitc As Byte
Dim en_tenc As Byte
Dim en_hunc As Byte
Dim en_thouc As Word
Dim en_totalc As Word
'START THE MAIN PROGRAM
solar_min_volt = 12600
batt_min_volt = 12100
energy = 0
'CLEAR SCREEN
LcdCmdout LcdClear
LcdCmdout LcdHome
LcdCmdout LcdLine1Home

'START MAIN PROGRAM
start:
Disable
Read 0, en_thou
Read 1, en_hun
Read 2, en_ten
Read 3, en_unit
Enable
en_total = en_thou * 1000 + en_hun * 100
+ en_ten * 10 + en_unit
Gosub chk_solar_volt
Gosub chk_batt_volt
If solar_volt > solar_min_volt Then
trickle_time_count = trickle_time_count
+ 1
If batt_volt < batt_min_volt Then
boost_on = 1
trickle_on = 0
EndIf
If batt_volt >= batt_min_volt Then
boost_on = 0
trickle_on = 1
EndIf
If boost_on = 1 Then
chg_on = 1
EndIf
If trickle_on = 1 Then
If trickle_time_count <= 2 Then
chg_on = 1
EndIf
If trickle_time_count > 2 Then
chg_on = 0
EndIf
If trickle_time_count >= 4 Then
trickle_time_count = 0
EndIf
EndIf
WaitMs 1000
Gosub chk_batt_curr
LcdCmdout LcdClear
LcdCmdout LcdHome
LcdCmdout LcdLine1Clear
LcdCmdout LcdLine1Home
Lcdout "B.Volts:", #batt_volt, " mV"
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
Lcdout "C.Curr:", #batt_curr, " mA"
WaitMs 2000

LcdCmdout LcdLine1Clear
LcdCmdout LcdLine1Home
Lcdout "Energy(Wh)", #en_total,
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
Lcdout "Power: ", #power, " mW"
WaitMs 2000
LcdCmdout LcdLine1Clear
LcdCmdout LcdLine1Home
Lcdout "Solar V:", #solar_volt, " mV"
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
Lcdout "B.Volt: ", #batt_volt, " mV"
WaitMs 2000
LcdCmdout LcdLine1Clear
LcdCmdout LcdLine1Home
If boost_on = 1 Then
Lcdout "Boost Chg: ON"
Else
Lcdout "Boost Chg: OFF"
EndIf
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
If trickle_on = 1 Then
Lcdout "Trickle Chg: ON"
Else
Lcdout "Trickle Chg: OFF"
EndIf
WaitMs 1000

Else
LcdCmdout LcdLine1Clear
LcdCmdout LcdLine1Home
Lcdout "Low Solar Volts"
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
Lcdout "Waiting..",
WaitMs 2000
LcdCmdout LcdLine2Clear
LcdCmdout LcdLine2Home
Lcdout "B.Volt: ", #batt_volt, " mV"
WaitMs 2000
EndIf
Goto start
End
'CHECK SOLAR CELL VOLTAGE
'return batt_volt in mV
chk_solar_volt:
If boost_on = 1 Then
was_boost = 1
Else
was_boost = 0
EndIf
If trickle_on = 1 Then
was_trickle = 1
Else
was_trickle = 0
EndIf
chg_on = 0
WaitMs 50
Adcin 2, solar_volt_raw
solar_volt = solar_volt_raw * 20
If was_boost = 1 Then
boost_on = 1
Else
boost_on = 0
EndIf
If was_trickle = 1 Then
trickle_on = 1
Else
trickle_on = 0
EndIf
Return
'CHECK BATTERY CURRENT
'return batt_curr in ma
chk_batt_curr:
For batt_curr_count = 0 To 4
Adcin 1, batt_curr_data(batt_curr_count)
WaitMs 10
Next batt_curr_count
batt_curr_raw = (batt_curr_data(0) +
batt_curr_data(1) + batt_curr_data(2) +
batt_curr_data(3) + batt_curr_data(4))
/ 5
batt_curr = batt_curr_raw
Return
'CHECK BATTERY VOLTAGE
'return batt_volt in mV
chk_batt_volt:
If boost_on = 1 Then
was_boost = 1
Else
was_boost = 0
EndIf
If trickle_on = 1 Then
was_trickle = 1
Else
was_trickle = 0
EndIf
chg_on = 0
WaitMs 50
Adcin 0, batt_volt_raw
batt_volt = batt_volt_raw * 20
If was_boost = 1 Then
boost_on = 1
Else
boost_on = 0
EndIf
If was_trickle = 1 Then
trickle_on = 1
Else
trickle_on = 0
EndIf
On Interrupt
Save System
en_count = en_count + 1
int_count = int_count + 1
'1 SEC INTERRUPT
If int_count = 15 Then
int_count = 0
power = (batt_volt / 1000) * (batt_curr)
energy_aux = energy_aux + power
'ENERGY IN WS
If energy_aux >= 1000 Then
energy = energy + 1
energy_aux = 0
EndIf
'ENERGY IN WH
If energy >= 3600 Then
en_totalc = en_totalc + 1
energy = 0
EndIf
If en_totalc >= 9999 Then
en_totalc = 0
EndIf
EndIf
'1/2 HR INTERRUPT
If en_count = 27000 Then
en_count = 0
en_unitc = en_totalc Mod 10
en_tenc = (en_totalc / 10) Mod 10
en_hunc = (en_totalc / 100) Mod 10
en_thouc = en_totalc / 1000
Write 0, en_thouc
Write 1, en_hunc
Write 2, en_tenc
Write 3, en_unitc
EndIf
INTCON.T0IF = 0
Resume
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